Banach Fixed-Point Theorem

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1]*Mindverse Computing LLC, Lynnwood, WA 98087, USA*

(Dated: October 2026)

Abstract

A complete authored walkthrough of banach fixed-point theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RA-03 / Real Analysis / Report v1

I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let (X, d) be a nonempty complete metric space and $T : X \to X$ satisfy $d(Tx, Ty) \leq qd(x, y)$ for all x, y, with a fixed $0 \leq q < 1$. Then there is a unique x_* with $Tx_* = x_*$. From any $x_0 \in X$, the sequence $x_{n+1} = Tx_n$ converges to x_*. For $0 < q < 1$,*

$$d(x_n, x_*) \leq \frac{q^n}{1-q} d(x_1, x_0).$$

The output is an existence proof, a uniqueness proof, a convergent construction, and an error certificate. An iterate with small consecutive displacement is an approximation; it is not automatically an exact fixed point.

A. Foundations and permitted imports

completeness means every Cauchy sequence in X has a limit in X; a contraction has one uniform factor q; geometric-series summation and the triangle inequality are imported. If $q = 0$, T is constant, so x_1 is already a fixed point. We prove the remaining case $0 < q < 1$.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space X.

Receiving activity. N2: Shrinking Increments and the Cauchy Certificate

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and the finite geometric-series bound.

T2: N2/E2/v1 to N3. $d(x_{n+p}, x_n) \leq q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

Receiving activity. N3: Completeness and the Fixed-Point Bridge

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2 and completeness of the original metric space.

T3: N1/E1/v1 to N4. For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space X.

Receiving activity. N4: Uniqueness

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1; two putative exact fixed points u, v.

T4: N2/E2/v1 to N5. $d(x_{n+p}, x_n) \leq q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

Receiving activity. N5: Synthesis and a Stopping Certificate

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N3/E3, and N4/E4.

T5: N3/E3/v1 to N5. There exists $x_* \in X$ with $T(x_*) = x_*$ and $d(x_n, x_*) \leq q^n d(x_1, x_0)/(1-q)$.

Receiving activity. N5: Synthesis and a Stopping Certificate

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N3/E3, and N4/E4.

T6: N4/E4/v1 to N5. Any two fixed points in X are equal.

Receiving activity. N5: Synthesis and a Stopping Certificate

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N3/E3, and N4/E4.

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: An Admissible Infinite Construction

Object and role. An Admissible Infinite Construction. Establish the following local result: For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space X.

Inputs. $T : X \to X$ and a uniform contraction constant $0 \leq q < 1$.

E — Ego

Own E1/v1: For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space X.

M — Memory

Metric and map definitions

I — Intellect

Choose any starting point

S — Sense

Check completeness, self-map, $q < 1$

A — Action

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Check nonemptiness, completeness, the self-map condition, and a single global constant q.

N1.2 If $q = 0$, the map is constant and its value is its unique fixed point.

N1.3 For $0 < q < 1$, fix any x_0 and form the iteration.

Export E1/v1. For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space X.

Acceptance gate. Check completeness, self-map, $q < 1$ Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Check completeness, self-map, $q < 1$ Recheck dependent nodes N2, N3, N4, N5.

B. N2/v1: Shrinking Increments and the Cauchy Certificate

Object and role. Shrinking Increments and the Cauchy Certificate. Establish the following local result: $d(x_{n+p}, x_n) \le q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

Inputs. N1/E1 and the finite geometric-series bound.

E — Ego

Own E2/v1: $d(x_{n+p}, x_n) \le q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

M — Memory

Geometric series

I — Intellect

Sum shrinking increments

S — Sense

Keep q uniform

A — Action

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Inductively prove $d(x_{n+1}, x_n) \le q^n d(x_1, x_0)$.

N2.2 Sum increments from n to $n+p$ using the triangle inequality.

N2.3 Let n grow while keeping the tail estimate uniform in p.

Export E2/v1. $d(x_{n+p}, x_n) \le q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

Acceptance gate. Keep q uniform Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Keep q uniform Recheck dependent nodes N3, N5.

C. N3/v1: Completeness and the Fixed-Point Bridge

Object and role. Completeness and the Fixed-Point Bridge. Establish the following local result: There exists $x_* \in X$ with $T(x_*) = x_*$ and $d(x_n, x_*) \le q^n d(x_1, x_0)/(1-q)$.

Inputs. N2/E2 and completeness of the original metric space.

E — Ego

Own E3/v1: There exists $x_* \in X$ with $T(x_*) = x_*$ and $d(x_n, x_*) \le q^n d(x_1, x_0)/(1-q)$.

M — Memory

Completeness

I — Intellect

Use contraction continuity

S — Sense

Keep limit in X

A — Action

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Take the Cauchy limit inside X.

N3.2 Use $d(Tx_*, x_*) \le q d(x_*, x_n) + d(x_{n+1}, x_*)$ and pass to the limit.

N3.3 Let the tail endpoint tend to infinity in the estimate.

Export E3/v1. There exists $x_* \in X$ with $T(x_*) = x_*$ and $d(x_n, x_*) \le q^n d(x_1, x_0)/(1-q)$.

Acceptance gate. Keep limit in X Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Keep limit in X Recheck dependent nodes N5.

D. N4/v1: Uniqueness

Object and role. Uniqueness. Establish the following local result: Any two fixed points in X are equal.

Inputs. N1/E1; two putative exact fixed points u, v.

E — Ego

Own E4/v1: Any two fixed points in X are equal.

M — Memory

Distance axioms

I — Intellect

Apply contraction to their distance

S — Sense

Compare exact fixed points

A — Action

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Apply the contraction inequality to u, v.

N4.2 Obtain $(1 - q)d(u, v) \leq 0$.

N4.3 Use $1 - q > 0$ and the metric separation axiom.

Export E4/v1. Any two fixed points in X are equal.

Acceptance gate. Compare exact fixed points Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Compare exact fixed points Recheck dependent nodes N5.

E. N5/v1: Synthesis and a Stopping Certificate

Object and role. Synthesis and a Stopping Certificate. Establish the following local result: The unique fixed point is reached in the limit from every starting point, with $d(x_n, x_*) \le d(x_n, x_{n+1})/(1-q)$.

Inputs. N2/E2, N3/E3, and N4/E4.

E — Ego

Own E5/v1: The unique fixed point is reached in the limit from every starting point, with $d(x_n, x_*) \le d(x_n, x_{n+1})/(1-q)$.

M — Memory

All predecessor certificates

I — Intellect

Join construction and bounds

S — Sense

Separate approximation from equality

A — Action

Carry out the numbered deductions and publish E5/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N5.1 Restart the geometric-tail estimate at index n.

N5.2 Use the observed next increment to produce a certified stopping tolerance.

N5.3 Combine existence, uniqueness, and both error estimates without equating a finite iterate to the exact limit.

Export E5/v1. The unique fixed point is reached in the limit from every starting point, with $d(x_n, x_*) \le d(x_n, x_{n+1})/(1-q)$.

Acceptance gate. Separate approximation from equality Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N5: Separate approximation from equality Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — An Admissible Infinite Construction

Choose $x_0 \in X$ and put $x_{n+1} = Tx_n$. Every iterate stays in X because T is a self-map. Set $D = d(x_1, x_0)$. If $D = 0$, the initial point is already fixed. Otherwise the following estimates still apply without changing the algorithm.

B. N2 — Shrinking Increments and the Cauchy Certificate

Induction using the contraction gives

$$d(x_{n+1}, x_n) \leq q^n D.$$

For $m > n$, the triangle inequality therefore gives

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} q^k D \leq \frac{q^n D}{1-q}.$$

The bound tends to zero independently of m. Given any epsilon, choose n large enough that the last expression is smaller than epsilon. Thus the sequence is Cauchy. This is a statement about every sufficiently late pair of iterates, not just adjacent ones.

C. N3 — Completeness and the Fixed-Point Bridge

Completeness supplies $x_* \in X$ with $x_n \to x_*$. Since

$$d(Tx_*, x_*) \leq d(Tx_*, Tx_n) + d(x_{n+1}, x_*) \leq qd(x_*, x_n) + d(x_{n+1}, x_*),$$

the right side tends to zero. Hence $Tx_* = x_*$. Letting m tend to infinity in N2's estimate gives the stated error bound. The continuity used here follows from the contraction inequality itself.

D. N4 — Uniqueness

If x and y are fixed points, then $d(x, y) = d(Tx, Ty) \leq qd(x, y)$. Consequently $(1 - q)d(x, y) \leq 0$. As $1 - q > 0$ and distance is nonnegative, $d(x, y) = 0$, so $x = y$.

E. N5 — Synthesis and a Stopping Certificate

Every initial point has a limit by N2–N3, and N4 shows all those limits are the same point. For a practical stopping criterion, N2 applied starting at x_n gives

$$d(x_n, x_*) \leq \frac{d(x_{n+1}, x_n)}{1-q}.$$

For $n \geq 1$ there is also the sharper bound

$$d(x_n, x_*) \leq \frac{q}{1-q} d(x_n, x_{n-1}).$$

Thus a requested tolerance can be met using observed increments and a proved contraction factor. This completes the theorem. □

□

V. WORKED ORCHESTRATION

Take $X = [0, 1]$ with ordinary distance and $T(x) = \cos$ x. The interval is complete, and T maps it into [cos 1,1], a subset of X. The mean value theorem gives $|\cos x - \cos y| \leq (\sin 1)|x - y|$. Use $q = \sin 1 \approx 0.841471 < 1$.

Starting from $x_0 = 0$:

- **n**

  0

  Approximate iterate

  0

- **n**

  1

  Approximate iterate

  1

- **n**

  2

  Approximate iterate

  0.540302306

- **n**

  3

  Approximate iterate

  0.857553216

- **n**

  4

  Approximate iterate

  0.654289790

- **n**

  5

  Approximate iterate

  0.793480359

The alternation does not invalidate convergence. A contraction bound controls its shrinking oscillations. After 100 iterations, the a priori error bound is $q^{100}/(1-q) < 2.1 \times 10^{-7}$. The limit is approximately 0.7390851332; this decimal is illustrative, while the theorem establishes the unique exact solution in [0,1].

A. Boundary and hypothesis audit

- **Boundary**

  $q = 0$

  Consequence

  T is constant; one application reaches the fixed point

- **Boundary**

  $q = 1$

  Consequence

  Identity on a multi-point space has many fixed points; uniqueness fails

- **Boundary**

  Missing completeness

  Consequence

  Limit may leave the space

- **Boundary**

  Map leaves X

  Consequence

  Iteration is not covered by the contract

- **Boundary**

  Finite decimal equality

  Consequence

  Does not certify an exact fixed point

Export the contraction theorem, uniqueness lemma, and both error estimates with their hypotheses. The Inverse Function Theorem example imports this completed module to solve nearby equations.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

Consider $T(x) = x/2$ on $X = (0, 1)$. A proposed branch claims a fixed point because the distances halve. Sense finds that X is not complete: the iterates converge to 0, which is missing.

- **Node:** N3
- **State:** STALLED
- **Failing faculty:** Sense
- **Failed hypothesis:** complete domain containing the limit
- **Example:** $T(x) = x/2$ on (0,1)
- **Repair target:** N1
- **Blocked export:** fixed-point existence in (0,1)

Memory retrieves completeness's definition. Sense retains the missing-point diagnosis. Intellect proposes [0,1] for this example. Action verifies that the repaired domain is complete and invariant and that 0 is fixed. Ego publishes a theorem for the repaired domain while recording that the original map on (0,1) has no fixed point. Repair does not retroactively prove the rejected original claim.

B. Hypothesis boundary

- **Boundary**

  $q = 0$

  Consequence

  T is constant; one application reaches the fixed point

- **Boundary**

  $q = 1$

  Consequence

  Identity on a multi-point space has many fixed points; uniqueness fails

- **Boundary**

  Missing completeness

  Consequence

  Limit may leave the space

- **Boundary**

  Map leaves X

  Consequence

  Iteration is not covered by the contract

- **Boundary**

  Finite decimal equality

  Consequence

  Does not certify an exact fixed point

Export the contraction theorem, uniqueness lemma, and both error estimates with their hypotheses. The Inverse Function Theorem example imports this completed module to solve nearby equations.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. University of Pennsylvania advanced-analysis material on contraction-based inversion.